

Irfan Soudagar, Ph.D.

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Operations research and decision analytics researcher specializing in optimization under uncertainty, with applications in maritime decarbonization, fleet planning, facility location and routing, and environmental policy analysis.

EDUCATION

National University of Singapore (NUS)

Aug 2016 – Aug 2021

Ph.D., Department of Analytics and Operations, NUS Business School

- GPA: 4.45/5; NUS Research Scholarship
- Thesis: *Data-Driven Robust Optimization of Facility Location and Routing Problems*

Indian Institute of Technology (IIT) Kharagpur

Aug 2009 – May 2014

Dual Degree: Bachelor's (Hons.) in Manufacturing Science & Eng., Master's in Industrial Eng. & Management

- GPA: 8.72/10; Institute Silver Medal for highest departmental GPA
- Best Master's Dissertation; IIT Kharagpur Graduate Fellowship

RESEARCH INTERESTS / EXPERTISE

- Optimization under Uncertainty, Data-Driven Decision Making, and Simulation–Optimization
- Maritime and Logistics: Planning, Operations, and Environmental Policy Analysis

SKILLS AND INTERESTS

Programming: Python, Pyomo, GUROBI, R, SQL, MATLAB, Microsoft Excel

Languages: English, Hindi, Telugu, Urdu ; **Interests:** Sports (Badminton, Tennis, Cricket), Travel

PUBLICATIONS

- (2026) Long He, **Irfan Soudagar**, and Zhang Yuli. “[Spatial-Data-Driven Facility Location Planning](#).” *Production and Operations Management*, 35(4), 1313-1332.
- (2023) Yonzheng J. Chua, **Irfan Soudagar**, Szu Hui Ng, and Qiang Meng. “[Impact Analysis of Environmental Policies on Shipping Fleet Planning Under Demand Uncertainty](#).” *Transportation Research Part D*, 120, 103744.
- (2022) **Irfan Soudagar** and Le Van Dang. “[An Artificial Neural Network for Production Planning Automation of Fast-Moving Consumer Goods](#).” *Proceedings of the 12th Conference on Learning Factories (CLF)*.
- (2017) Ravi Shankar Kumar, Alok Choudhary, **Irfan Soudagar**, Sri Krishna Kumar, A. Goswami, and Manoj K. Tiwari. “[Designing Multi-Period Supply Chain Network Considering Risk and Emission: A Multi-Objective Approach](#).” *Annals of Operations Research*, 250, 427–461.
- (2015) **Irfan Soudagar**, Saurabh Pratap, Geet Lahoti, Kiran J. Fernandes, Manoj K. Tiwari, Matthew Mount, and Yu Xiong. “[Minimizing Delay of Ships in Bulk Terminals by Simultaneous Ship Scheduling, Stockyard Planning and Train Scheduling](#).” *Maritime Economics and Logistics*, 17, 464-492.

WORKING / DISCUSSION PAPERS

- (2026) **Irfan Soudagar** and Szu Hui Ng. “[Navigating Environmental Compliance: Maritime Fleet Renewal Under Fuel Market Uncertainty](#).” Under review at *Transportation Research Part D*.
- (2025) Iman Jafari, **Irfan Soudagar**, and Szu Hui Ng. “[Navigating IMO Net-Zero Framework: Potential Challenges and Refinements](#).” Discussion Paper available at *NUS Centre for Maritime Studies*.

RESEARCH AND PROFESSIONAL EXPERIENCE

Senior Research Fellow | National University of Singapore (NUS)

Jan 2025 – Present

Global Maritime Fuel and Technology Transition Modelling

- Develop a simulation–optimization model of global maritime fuel and technology uptake, incorporating fuel switching, retrofits, newbuilds, and energy-efficiency measures under technical, cost, fuel-availability, and regulatory constraints.
- Capture dynamic cost–uptake feedbacks to project fleet composition, fuel mix, and emissions pathways.

Project Lead – Economic Value and Impacts of Maritime Decarbonisation Measures on Singapore

- Lead quantitative analysis of fuel rewards, compliance markets, and ship compliance and competitiveness under global and fragmented regulatory scenarios.
- Lead project coordination across research teams, integrating analytical inputs, timelines, and deliverables.

Maritime Fleet Renewal Planning: Navigating Environmental Compliance under Fuel Cost and Supply Uncertainty

- Model fleet renewal under uncertainty in low-carbon fuel costs and supply, and assess how alternative regulatory designs (e.g., IMO Net-Zero Framework, levy) affect investment and operational compliance decisions.
- **Outcome:** SMI-funded research; paper under review at *Transportation Research Part D*.

Project Lead – Impact Analysis of Maritime Decarbonisation Mid-Term Measures

- Developed a quantitative assessment model to evaluate candidate maritime decarbonisation measures across multiple fuel-transition pathways.
- Quantified impacts on fuel costs, compliance-unit supply and demand, revenues, and fuel rewards across alternative regulatory designs and fuel-transition scenarios.
- **Outcome:** Supported MPA at IMO MEPC 83 (London) in assessing key design parameters including carbon pricing, fuel-intensity targets, and revenue allocation.

Research Fellow | National University of Singapore (NUS)

Jul 2022 – Dec 2024

Project Lead – Singapore-Rotterdam Maritime Green Corridor: Cost Gap Modelling and Analysis

- Developed a long-term fleet-renewal model integrating alternative-fuel vessels, energy-efficiency retrofits, fuel costs and availability, trade demand, and decarbonisation requirements.
- Quantified the cost gap to meet corridor decarbonisation targets and derived implications for future fuel mix and fleet composition to inform cost-sharing and regulatory options.
- Coordinated end-to-end project delivery with stakeholders (port authorities, liner operators, and fuel suppliers).
- **Outcome:** Presented findings at project consortium workshops in Rotterdam and Singapore.

Impact Analysis of ETS on Maritime Fleet Deployment Operations

- Analyzed how Emissions Trading Systems (ETS) affect liner-shipping fleet deployment and emissions outcomes.
- Compared global and regional ETS designs under alternative carbon-price and emissions-cap settings to evaluate effectiveness and stakeholder impacts. **Outcome:** Published in *Transportation Research Part D*.

Decarbonization Analytics and Intelligence

- Coordinated cross-functional proposal development and built working knowledge of carbon accounting (Scopes 1–3), decarbonisation pathways, and implementation challenges across industries.

Research Scientist | A*STAR, Singapore

Oct 2021 – Jun 2022

Technical Lead – Modelling, Simulation and Optimization of Hyper-Personalized Manufacturing

- Developed optimization algorithms for production scheduling in a hyper-personalized manufacturing system.
- Coordinated a cross-functional team delivering simulation and optimization solutions.

PhD Researcher | National University of Singapore (NUS) Aug 2016 – Aug 2021

- Developed distributionally robust optimization models for facility location under continuously distributed, non-stationary spatial demand, with applications to emergency response.
- Formulated data-driven robust optimization methods for pickup-and-delivery routing with time windows under uncertain travel times.

Senior Associate | Think Analytics India (Think360.ai), India Nov 2015 – Jul 2016

- Developed pharmaceutical analytics and contract break-even models using SQL, Excel, and VBA to support pharmaceutical manufacturer contracting strategies.

Associate | Axtria India Private Limited, India Jul 2014 – Oct 2015

- Integrated pharmaceutical sales and physician data using SQL/SSIS and developed performance dashboards and analytics for client decision-making.

Master's Researcher | IIT Kharagpur, India Jan 2013 – May 2014

- Designed a multi-objective optimization framework for multi-tier, multi-period supply chain networks incorporating inter-tier social relationships. **Outcome:** Published in *Annals of Operations Research*.
- Developed an integrated optimization/heuristic framework for ship scheduling, stockyard planning, and train scheduling to reduce vessel delays. **Outcome:** Published in *Maritime Economics and Logistics*.

Undergraduate Researcher | University of York, UK May 2013 – July 2013

- Developed a fuzzy-ANP expert system for project selection in open process innovation.
- Evaluated pricing strategies under competitive bidding in the UK road-haulage industry.

TEACHING EXPERIENCE | National University of Singapore

- *Decarbonisation: Principles, Metrics and Cases* – Guest Lecturer, AY2025/26
- *Decision Analytics Using Spreadsheets* – Lecturer, AY2018/19 and AY2020/21; Tutor, AY2017/18
- *Programming for Business Analytics* – Tutor, AY2018/19

GRANTS AND RESEARCH PROPOSALS

- (2026) “*Economic Value and Impact of Maritime Decarbonisation Measures on Singapore.*” Co-developed research concept and grant proposal; funded by Singapore Maritime Institute (SMI).
- (2024) “*Decarbonisation Strategies for Liner Company Fleet Renewal: Navigating Uncertainty in Alternative Fuels Cost and Supply.*” Co-developed research concept and grant proposal; funded by Singapore Maritime Institute (SMI).

SELECTED PRESENTATIONS

- (2025) “The Maritime Decarbonisation Journey,” Centre for Maritime Studies 20th Anniversary, NUS, Singapore.
- (2025) Technical support to the Maritime and Port Authority of Singapore (MPA) on “Impact Assessment of IMO Mid-term Measures,” MEPC 83, London.
- (2023, 2024) “Cost-Gap Modelling for Singapore–Rotterdam Green Shipping Corridor,” Singapore Maritime Week (SMW).
- (2022) “An Artificial Neural Network for Production Planning Automation of Fast-Moving Consumer Goods,” 12th Conference on Learning Factories (CLF), Singapore.
- (2019) “Robust Location Problems Under Spatial Uncertainty,” INFORMS Annual Meeting, Seattle, USA.